Applications: Growth and Decay

Section 1: Population Growth Models

1 A bacterial culture doubles every 3 hours. If the initial count is 500, what is the population after 9 hours?

a. 4000

b. 1500

c. 2000

d. 8000

2 A city has population $P(t) = 100000e^{0.02t}$, where t is in years. Find the continuous annual growth rate and doubling time.

a. Rate $= 2\%$, Doubling time ≈ 34.7 years

b. Rate $= 20\%$, Doubling time ≈ 3.5 years

c. Rate $= 0.02\%$, Doubling time ≈ 50 years

d. Rate $= 2\%$, Doubling time ≈ 50 years

3 If a culture grows from 200 to 600 in 4 hours according to $P(t) = P_0e^{kt}$, find k.

a. $k = \frac{\ln 3}{4} \approx 0.275$

b. $k = \frac{\ln 4}{3}$

c. $k = 3/4$

d. $k = \ln(3/4)$

Section 2: Radioactive Decay and Half-Life

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4 Carbon-14 has a half-life of 5730 years. What fraction of the original C-14 remains in a fossil after 17190 years?

a. 1/8

b. 1/4

c. 1/16

d. 1/3

5 State the formula relating decay constant $k > 0$ and half-life $t_{1/2}$ in $N(t) = N_0 e^{-kt}$.

a. $k = \frac{\ln 2}{t_{1/2}}$

b. $k = \frac{t_{1/2}}{\ln 2}$

c. $k = \frac{1}{2t_{1/2}}$

d. $k = \ln(2t_{1/2})$

6 A sample contains 80 mg of an isotope with a half-life of 10 days. How much remains after 25 days?

a. $80(0.5)^{2.5} \approx 14.14$ mg

b. 20 mg

c. 10 mg

d. 15 mg

Section 3: Newton's Law of Cooling

7 State Newton's Law of Cooling formula for temperature $T(t)$ with ambient medium T_m.

a. $T(t) = T_m + (T_0 - T_m)e^{-kt}$

b. $T(t) = T_0 e^{-kt} + T_m$

c. $T(t) = T_m(1 - e^{-kt})$

d. $T(t) = (T_0 - T_m)e^{kt}$

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8 A hot cup of coffee at $190\,^\circ\text{F}$ is placed in a $70\,^\circ\text{F}$ room. What is the initial temperature difference $T_0 - T_m$?

a. $120\,^\circ\text{F}$

b. $190\,^\circ\text{F}$

c. $70\,^\circ\text{F}$

d. $260\,^\circ\text{F}$

9 If the coffee cools from $190\,^\circ\text{F}$ to $130\,^\circ\text{F}$ in 10 minutes in the $70\,^\circ\text{F}$ room, find k.

a. $k = \frac{\ln 2}{10} \approx 0.0693 \text{ min}^{-1}$

b. $k = 0.1$

c. $k = \frac{\ln 3}{10}$

d. $k = 0.05$

Section 4: Logarithmic Scales (pH, Richter, Decibels)

10 The pH of a solution is defined as $\text{pH} = -\log_{10}[\text{H}^+]$. What is the pH if $[\text{H}^+] = 10^{-4}$ M?

a. 4

b. -4

c. 10

d. 14

11 The Richter magnitude scale is $M = \log_{10}\left(\frac{I}{S}\right)$. How many times more intense is an earthquake of magnitude 7 than one of magnitude 4?

a. 1000 times (10^3)

b. 3 times

c. 30 times

d. 10000 times

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12 Sound intensity level in decibels is $\beta = 10\log_{10}\left(\frac{I}{I_0}\right)$. If the intensity I is multiplied by 100, how many decibels are added?

a. +20 dB

b. +100 dB

c. +10 dB

d. +2 dB

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